

Linux Programming: Assignment-1

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Section: A

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Q1. What is Linux Operating System (OS)? List three pros and cons of it.

Linux is an open-source operating system on the Unix architecture. It controls hardware and offers an environment for running applications.

Pros:

- Open-source & free
- Very secure with less viruses
- Stable & trustworthy for servers

Cons:

- More difficult for new users
- Software compatibility issues by some applications
- Limitations in hardware drivers

Q2. Compare Linux, Mac, Android, and Windows OS with at least six distinguishing features.

Linux: Open source, customizable, free, secure, utilized in servers.

MacOS: Apple's closed-source, costly, highly secure, creative professionals.

Android: Linux kernel-based, for mobiles, customizable, large user base.

Windows: Closed source, costly, common usage in homes/offices, more vulnerable to malware.

Q3. Why is Linux used in Mainframe Servers for legacy application?

- Stability and dependability
- Backward compatibility
- Scalability and performance

Q4. Describe the Linux File System with proper diagram.

Linux filesystem is a tree filesystem with root '/' at top. Important directories:

- /bin: must have binaries
- /etc: configuration
- /home: users' files
- /var: logs
- /lib: libraries
- /boot: boot files
- /usr: programs

Diagram: / → bin, boot, etc, home → user, lib, usr, var

Q5. If Linux OS is open-source, then how are companies like Red Hat still making money out of it?

- Enterprise support subscriptions
- Consulting services
- Training & certifications
- Partnerships for enterprise-ready Linux solutions

Q6. What is the command to print today's date and time?

Command: date

Q7. What command is used to see how long the system has been running?

Command: uptime

Q8. How is shutdown -h now different from halt?

shutdown -h now: Terminates processes and unmounts filesystems gracefully.
halt: Stops CPU immediately, may not safely unmount. Shutdown is better.

Q9. What is the difference between init 0 and shutdown -h? Which one is safer? Why?

init 0: Changes runlevel to 0 (shuts down system).

shutdown -h: Shuts down system gracefully.

Safer: shutdown -h since it guarantees cleanup.

Q10. A system administrator inadvertently shuts down a Server machine using the power button instead of gracefully shutting it down. What can go wrong?

- File corruption
- Loss of unsaved configurations
- Filesystem damage
- Service downtime